Multi-class classification of song emotions using Machine learning

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***Abstract*—Recent developments in Music Information Retrieval (MIR) have importantly brought several opportunities to enhance those applications that require music emotion recognition. Combining the terms 'music', 'emotion,' and 'recognition' firstly triggers the meaning that it is the detection of audio emotion in a musical form. The work in this paper uses the Music4All dataset and explores the possibility of audio metadata features-based music emotion recognition. The audio features, 13 in number, were initially analyzed to determine the emotion label. The audio emotion recognition task discussed in this paper is based on Russell's Circumplex model, which examines four classes of music emotion. The features which are relevant according to this model are valence and energy (same as arousal feature). Based on the range of values of valence and energy, three classifiers were built: Support Vector Machine (SVM), Logistic regression, and artificial neural networks (ANN). Logistic regression performed significantly better than the deep learning model by generating 99.9% accuracy in multi-class audio classification.**

Keywords— Music emotion recognition, multi-class classification, emotionally relevant features, music recommender systems, Machine Learning, Neural Networks and Deep Learning

I. INTRODUCTION

Due to the growth of the internet and easy access to multimedia information, digital music has become more readily available to everyone. With the ease of access and affordable price of music, the number of listeners or audiences interested in music has become drastically increased over time. Multimedia streaming applications became the primary industry of entertainment where music information retrieval research has done many more advancements from automated methods for searching the music and performing several music-related tasks. With the proliferation of large, widely available digital music archives, music information retrieval research also had extensive growth with diverse attractive areas of interest [10].

Music information-based research on most music datasets is significant in identifying artists or genres. Music is popularly known as the "language of emotion," which means music brings an emotional state that invokes certain feelings [1, 3, 10] and interpretation and is harder to assess. Automatic emotion detection can be performed by the application of machine learning models. Recognition of the emotional content of music audio could can be carried out through multiple angles by applying these machine learning models. Music information retrieval is a multidisciplinary endeavour that includes the learning of psychology, auditory perception, audio signal processing, and machine learning [2, 10]. Automatic emotion recognition focuses on computational methods for music mood recognition using a psychological model of emotion [5, 6, 7]. The emotion models for music are categorical or non-categorical. As more and more advancements have happened in machine learning, music emotion detection works need the most acceptable ways to predict the emotion labels from scratch. The process of the Music Emotion Recognition (MER) methods is generally implemented based on genre tags, the artist, language, album, etc. MER has become a crucial part of many applications such as playlist generation and music recommender systems.

Music Emotion Detection Process

The Music emotion detection process starts with first understanding emotion annotations in the dataset [28]. Once the emotion labels are identified, further, emotionally relevant features need to be identified. The emotion labels and the features are considered training inputs [28]. After that music emotion prediction model is built. This model will further understand the extracted features of the unknown songs and then used them to identify their emotion. Further, the final model can be used as an emotion classifier for unseen music [28].

This work first provides an overview of some of the psychological studies utilized in developing machine learning models for automatic emotion recognition of music. Emotion annotation of music is basically labelling the music dataset. Diverse approaches for labelling emotion with promising solutions, such as listener surveys, crowdsourcing, social tags, and data collection games, are discussed in [11]. However, because emotion is not fully encompassed by audio alone, techniques involving music metadata, such as tags and lyrics, are also examined in depth [10]. This paper extends the binary emotion classification work on the same dataset, where a promising approach to emotion detection in two emotions, happy and sad, was discussed [12].

II. LITERATURE REVIEW

Before discussing automatic music emotion computational models built on machine learning algorithms, it is very crucial to discuss the psychological models which are the fundamental for these models [8, 15- 20]. One of the important categorical models in the literature is described in [8], where the author Ekman suggests that the mood of the music is purely related to the language used to represent it. Panksepp [9, 13] suggests seven categories of primal emotions that invoke human beings' positive or negative triggering stimuli. Emotions that trigger positive emotions are categorized as care, seeking, play, and lust, and negative emotions are categorized as sadness, anger, and fear [9, 13]. Thayer's two-dimensional model suggests that music energy and music pleasance are important parameters relevant to identifying the

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emotions from music [14]. These parameters are the fundamental basis of Thayer's model, where the music energy is represented as arousal, and music pleasance is defined as valence. Music pleasance is a parameter related to the cortisol levels in the human blood, and high valence corresponds to a high level of cortisol [15]. Highly energetic music will have increased energy, significantly increasing the audiences' heartbeat and blood pressure levels [15].

Russell's Circumplex model of affect is represented using the two-dimensional space of Thayer's model is modeled as a circular emotional plane with arousal on the vertical axis and valence on the horizontal axis [4]. When both axes join at the center, the valence will be neutral, and the arousal will be less or medium. Russell proposes that multiple emotional states can be represented using the diverse range of arousal and valence. The Russel Circumplex model was employed in this study to classify emotions according to music pleasance and arousal. As music energy is the essential concept behind the parameter used in Thayer's and Russell's model [14, 4], the attribute "energy," related to the arousal parameter, is selected for this work. According to the model (depicted in Figure 1), emotional expressivity is calculated: If both energy and valence values are positive, the emotion falls under the first happy quadrant. If energy is positive and valence is negative, the emotion is anger. The emotion falls under the sad third quadrant if both energy and valence values are negative. If the valence is positive and the energy is negative, the emotion is relaxed.

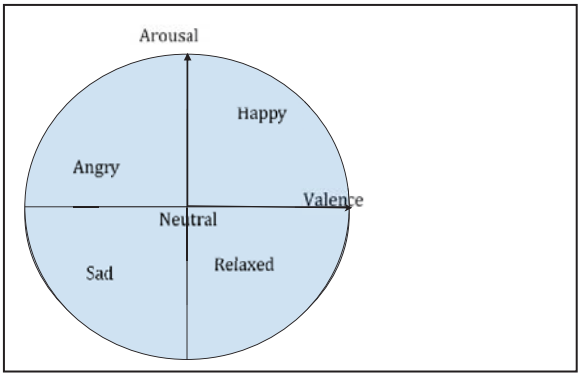

Fig. 1. Russell's Circumplex emotional plane [4]

Related works

This section describes various machine learning approaches used to detect music emotion [15-20]. In [16], the authors proposed a music emotion recognition approach that identified music mood from multiple music features and created a dataset based on audio, lyrics and combined data of the song. In the paper [17], the authors performed multi-class emotion classification to identify the mood of the music according to different mood categories in the Tellegen-Watson-Clark model. In [18], the authors achieved emotion recognition based on physiological signals rather than audio-video emotion channels and named the Emotion-specific multilevel dichotomous classification (EMDC) model. The authors considered different features that can change facial expression or speech and performed several studies utilizing time, frequency, multi-scale entropy, and geometric analysis [18]. The importance of audio features and social media user annotations was analyzed and applied to perform music mood classification [19]. Audio embeddings were exploited to study the music emotion by representing it through emotion semantics [20]. The work also suggests that deep audio embeddings can help find music audio emotion without the need for any manual feature engineering approach. In [21], authors performed emotion classification based on genre annotations from an online repository. In [21], the emotion classification research has conducted a study on multi-linguistic songs and studied the music with the same emotion from multiple languages and their different features.

Acoustic features like Jazz vocal and Classical sound were studied to find the music similarity and emotion [22]. The audio emotion classification was diversely explored in [23] based on the temporal, loudness, harmonic and rhythmic feature extraction to achieve a powerful automatic emotion detection model. The authors of [24] studied different music genres and their influence on several musical emotions. The work in this paper is purely based on the audio-metadata features available in the Music4All dataset, which were extracted from the Spotify music streaming application [27]. In [25], the authors identified music emotions based on the text extraction from the song lyrics.

III. DATASET

The Music4All dataset contains multi-modal information of more than one lakh songs extracted from the Spotify app [27]. The information given as the song metadata information includes Music4All_id, Spotify id, popularity, release, danceability, energy, key, mode, valence, tempo, and duration in milliseconds. The number of rows used in this work is 109269 in number.

The data preparation

The multi-class classification task needs to prepare the audio metadata according to the Psychological dimensional models discussed in the literature review section. Data should be categorized according to Russell's model, widely used in the literature in this work. The attributes which are according to the parameters of the valence-arousal plane used in Thayer's model and Russell's model in the Music4All dataset is the valence column and energy column [27]. The values in these two columns range from 0 to 1. The values were mapped into two categories a) values>=0 and <0.5 are taken as negative, b) values>=0.5 and <=1 to normalize the values according to the positive-negative values of Russell's plane. The valence (v) values will be determined based on the following range [26]:

$$0 <= v <= 1 \tag{1}$$

The energy (e) values will be determined based on the following range [26]:

$$0 <= e <= 1 \tag{2}$$

Next, the values in these categories are mapped to four quadrants, as shown in Table 1.

TABLE I. THE MAPPING OF THE EMOTIONAL ATTRIBUTES ACCORDING TO RUSSELL'S CIRCUMPLEX MODEL

Range	*quadrant*	*class*	*Emotional quadrant*

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Energy >= 0.5 and valence >=0.5	I	0	Happy
Energy >= 0.5 and valence < 0.5	II	1	Angry
Energy<0.5 and valence <0.5	III	2	Sad
Energy<0.5 and valence >=0.5	IV	3	Relaxed

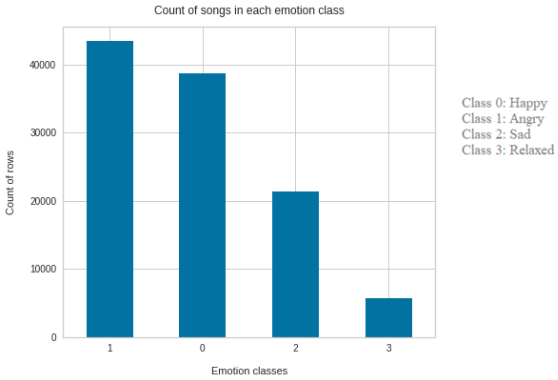

Fig. 2. The distribution of labels in each emotion class.

After categorizing the valence-energy values into four classes, the next step was to label the classes for each song used in this work and then find the distribution of each class in the dataset. Figure 2 shows the count of each class of songs in the Music4All dataset [27]. As seen in figure 2, the number of songs in the class angry is the highest, which is 43445 in the count. The second-highest number of songs is in the class happy, which is 38750 in number. The number of songs in the sad quadrant is 21318, and the least number of songs appear in class relaxed.

Feature extraction

The next step in this work was to identify more similar features related to the fundamental parameters of Russell's model. A correlation check is required to perform whether more features can contribute to the emotion detection task for this work. For that, the Pearson correlation method was applied to the data. It was found that the attributes of danceability and popularity are also correlated to the valence and energy column. The Pearson's correlation matrix is shown in the figure 3:

	popularity	release	danceability	energy	key	mode	valence	tempo	duration_ms
popularity	1.00	0.00	0.24	-0.02	-0.01	0.03	0.14	-0.01	-0.07
release	0.00	1.00	0.06	0.11	0.00	-0.07	-0.17	0.02	-0.03
danceability	0.24	0.06	1.00	-0.01	0.01	-0.03	0.52	-0.13	-0.16
energy	-0.02	0.11	-0.01	1.00	0.02	-0.03	0.26	0.23	-0.01
key	-0.01	0.00	0.01	0.02	1.00	-0.15	0.02	-0.00	0.01
mode	0.03	-0.07	-0.03	-0.03	-0.15	1.00	0.02	0.01	-0.05
valence	0.14	-0.17	0.52	0.26	0.02	0.02	1.00	0.07	-0.22
tempo	-0.01	0.02	-0.13	0.23	-0.00	0.01	0.07	1.00	-0.02
duration_ms	-0.07	-0.03	-0.16	-0.01	0.01	-0.05	-0.22	-0.02	1.00

Fig. 3. Pearson correlation matrix

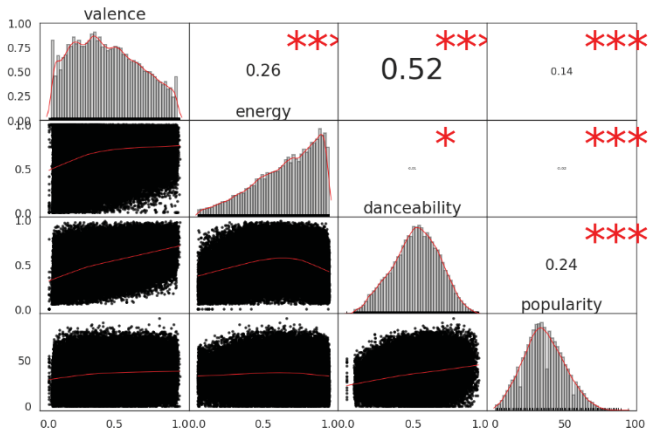

Fig. 4. Pair plots for correlated features

The plot in the first row and first column of figure 4 shows the distribution of valence values in the dataset. As shown in figure 4, the valence and energy columns show a positive correlation of 0.26. Additionally, there is a positive correlation of 0.52 between the valence and danceability columns. A positive correlation of 0.24 exists between the danceability and popularity columns. The valence and popularity columns exhibit a positive variance of 0.14. Other correlations have low values, indicating a low degree of correlation. Moreover, this means a weak correlation between the valence and energy columns. At the same time, valence and danceability features are strongly correlated. The danceability and popularity features have a weak correlation with each other. Even though the valence and popularity columns show a positive correlation, they exhibit a very low correlation among them.

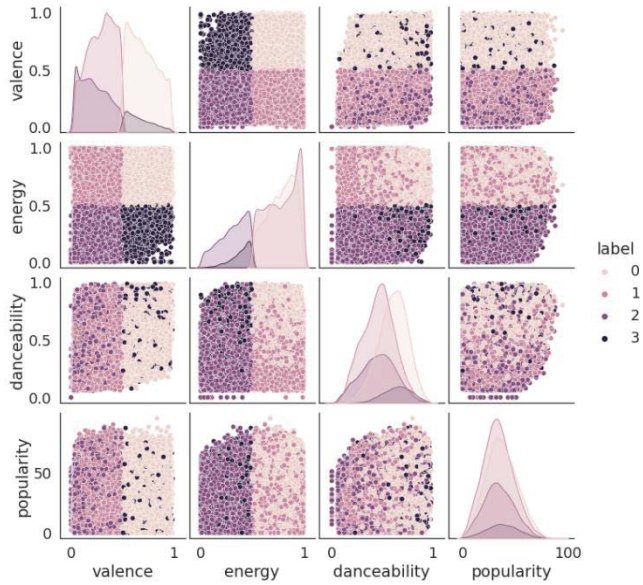

Fig. 5. Pair plot graph with label column as heuristics for all input features and classes

Figure 5 shows the distribution of each class and its corresponding features. The density and scatter plots are combined and displayed as the pair plots illustrated in figure

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5. The density plots show the smooth distribution of a data column. The scatter plot plots the relationship between two features, which helps understand the two features' relationship. The first plot in the first row and column shows the density plots of the valence column along the x-axis, valence-energy, valence-danceability, and valence-popularity features in the x-y axes. The valence-energy values are equally distributed for all emotion classes. The valence-danceability and valence-popularity plots are randomly distributed for emotion classes sad and relaxed. At the same time, the plots for classes happy and angry are equally distributed. The reason is the valence appears to be positively associated with happy music and negatively related to angry music. Additionally, valence is strongly associated with danceability, as both happy and angry songs can be considered danceable. Also, the number of songs in the sad(21318) and relaxed(5756) categories is comparatively low.

The density plots are plotted in energy-valence, energy-energy, energy-danceability, and energy-popularity. The second row of the plot shows the relationship between energy and other columns. The third row shows the relation between danceability and other columns such as valence, energy, danceability, and popularity. The last row shows the relation between popularity and other columns such as valence, energy, danceability, and popularity. The first graph in the first row of figure 6 shows that the density of valence is distributed between 0.5 to 0.1 for class 0 or 3 on the y-coordinate, which denotes happy or relaxed emotion. For classes 1 and 2, which represent angry or sad, the distribution of valence values lies between 0 to 0.5. Random distributions of minor classes severely impact the distribution of emotion labels in the dataset.

Multi-class classification approach

The Support Vector Machine (SVM) algorithm is a promising technique used in classification and used in different information retrieval tasks and data mining, machine learning, and predictive analytics. To build the SVM classifier on the data, initially, the dataset was divided into 7:3 training-test ratios. The extracted features are the input features to feed the classifiers to perform multi-class classification. Below in figure 6, the results of the SVM approach are shown.

	precision	recall	f1-score	support
0	0.998882	0.997252	0.998066	11644
1	0.995934	0.999000	0.997464	12994
2	1.000000	0.992396	0.996184	6444
3	0.983787	1.000000	0.991827	1699
accuracy	0.997132	0.997132	0.997132	0
macro avg	0.994651	0.997162	0.995885	32781
weighted avg	0.997151	0.997132	0.997134	32781

Fig 6: Multi-class SVM classification report.

The average and micro accuracy of the SVM model is 99.7 percent. The precision-recall, f1-score and support for classes 0,1, 2, 3 (happy, sad, angry and relaxed) are (99.8, 99.7, 99.8 and 11644), (99.5, 99.9, 99.7 and 12994) , (100, 99.2, 99.6, 6444) and (98.3, 100, 99.18 and 1699) respectively.

The graph plotted in figure 7 shows the learning rates of the SVM model.

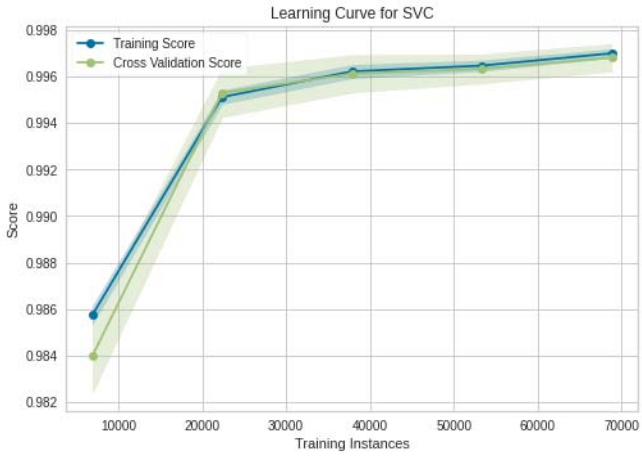

Fig 7: The learning curve of the SVM model

In figure 7, the training accuracy increase after 25000 instances. The training curve steadily increases from 10000 to 25000 training instances and gives 98.6 to 99.4 percent accuracy. After 25000 instances, the accuracy gradually increases and reaches the topmost accuracy of 99.8 percent after 68000 instances. In figure 7, the training accuracy increase after 25000 instances. The training curve steadily increases from 10000 instances to 25000 instances, giving 98.6 to 99.4 percent accuracy. After 25000 instances, the accuracy gradually increases and reaches the topmost accuracy of 99.8 percent after 68000 instances.

The Dense neural network model

The following approach is used to build a dense neural network. There are four layers in this model. The input layer consists of the dense layer. The second layer, the hidden layer, is the drop put layer, which prevents the model from overfitting. The third layer is again dense, and the last layer is the output layer which predicts the emotion label as shown below

Model:"sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	320
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 16)	1040
dense_2 (Dense)	(None, 4)	68

Total params: 1,428
Trainable params: 1,428
Non-trainable params: 0

The average accuracy of the deep learning model is 84.5

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The training and validation process of the model is plotted in figure 8.

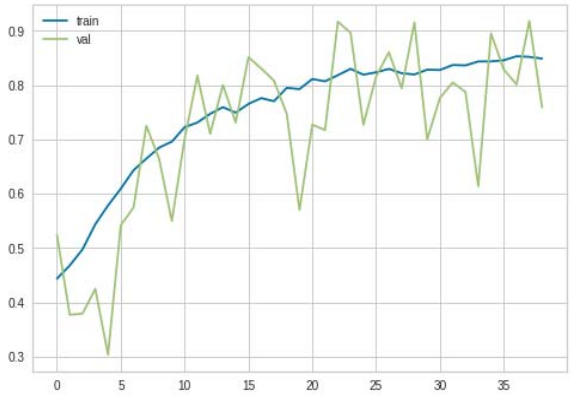

Fig 8: Training epochs.

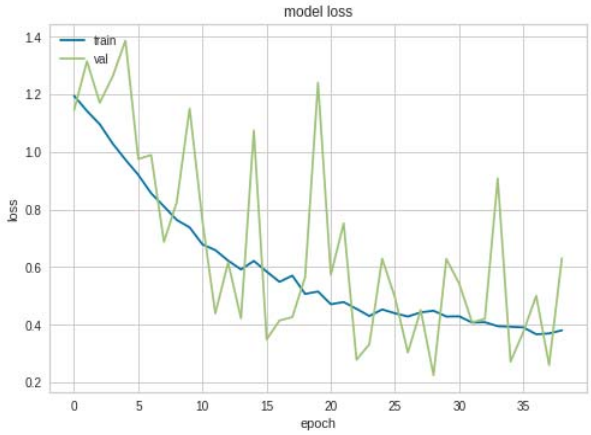

Fig 9: Model loss of dense neural network

The evaluation loss and accuracy: - loss: 0.2221, accuracy: 0.9203. The minimum loss is between the 35th and 40th epoch. The training stops at the 36th epoch as maximum accuracy is achieved with at a minimum loss.

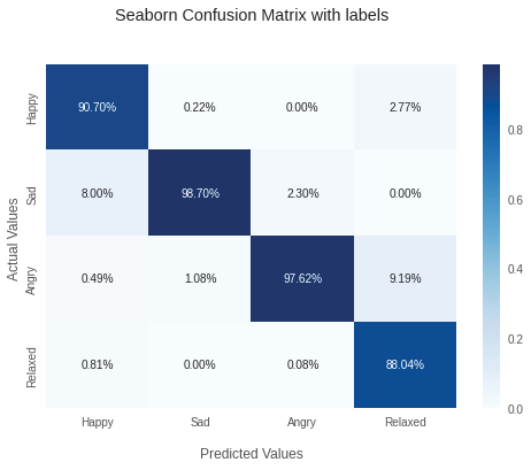

Fig 10: The confusion matrix for the deep learning model.

The confusion matrix the figure 10 shows that around 90.7 percent of happy songs are predicted as happy(true positive), 98.7 percent of sad songs are expected as sad, 97.64 percent of angry songs are predicted as angry, and 88.0 percent cent of relaxed songs are accurately predicted as relaxed. The above confusion matrix shows that around 90.7 percent of happy songs are happy (true positive), 98.7 percent of sad songs are sad, 97.64 percent of angry songs are predicted as angry, and 88.0 percent of relaxed songs are accurately predicted as relaxed. Among the given emotion classes, relaxed achieved the lowest true positive rate. This can be because the number of relaxed songs available in the dataset is comparatively low(approximately 6000 samples) compared to other emotion classes(more than 20000 samples). Among the given emotion classes, relaxed achieved the lowest true positive rate. This can be because the count of relaxed songs available in the dataset is comparatively low (approximately 6000 samples) compared to other emotion classes (more than 20000 samples).

The Logistic regression model

The logistic regression model was applied to the dataset, and the following results were obtained.

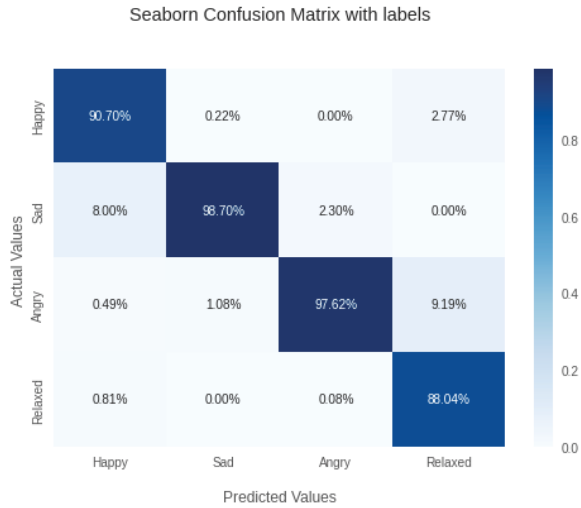

Fig 11: Confusion matrix for multinomial logistic regression

The class with high accuracy is happy and the least accuracy is relaxed [Figure 11, 12]. The micro average accuracy for logistic regression is 99.9. Figure 12 shows that the precision-recall and f1-score are 99.9 percent for logistic regression.

	precision	recall	f1-score	support
0	1.000000	1.000000	1.000000	11570
1	0.998642	1.000000	0.999321	13239
2	0.998402	0.997127	0.997764	6266
3	1.000000	0.994138	0.997061	1706
accuracy	0.999146	0.999146	0.999146	0
macro avg	0.999261	0.997816	0.998536	32781
weighted avg	0.999146	0.999146	0.999145	32781

Fig 12: The classification report of logistic regression

CONCLUSION AND DISCUSSION

The multi-class classification approaches applied to the Music4All dataset showed good accuracy results [27]. The logistic regression got high accuracy of 99.9, and precision

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and recall for all emotion classes are nearly 99-100%, as shown in figure 12. The Music4All dataset's meta-data information is well suited for emotion classification [27]. Applying different models such as SVM, Logistic regression, and dense neural network, a state-of-the-art logistic regression model achieved 99.9 percent accuracy. The neural network type architecture showed 82% percent of accuracy in predicting emotions. This work can be further extended by finding more emotions from the dataset. In this work, emotionally relevant features were identified, and then a detailed study on their distribution in each class type. The multi-class classification based on audio features implemented in this work guarantees that if contextually relevant features are correctly identified, a standard set of algorithms are enough to perform multi-class classification on any dataset under any application domain. This work can be applied to generate relevant music recommendations based on emotion.

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